

Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Outperform Optical and ImageNet Transfer Across ViT and CNN?

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Abstract

Detecting dark vessels—vessels lacking matched Automatic Identification System reports—in spaceborne synthetic aperture radar (SAR) supports maritime safety and enforcement, but human-verified examples are scarce. We study whether source-domain pretraining improves label efficiency on xView3-SAR, and whether the effect persists across architectures. Two size-matched tracks (ViT-B/16 and ConvNeXt-V2-Base) each compare random initialization with released ImageNet, optical remote-sensing, and SAR checkpoints under one detector, optimizer, schedule, scene-level split, and seed. Fine-tuning nested 10–100% fractions of 111 training scenes yields 32 controlled cells, all completed on H100 GPUs in strict FP32 and re-derived at build time from hash-bound evidence. Transfer helps most under label scarcity, but the SAR-versus-optical contrast is architecture-dependent: SAR pretraining dominates optical in the CNN track at every budget while the ViT track prefers the optical checkpoint below half data, and generic ImageNet initialization leads both tracks at full data. Reported values are corrected development-selection F1. The 50-scene human-verified evaluation remains sealed for follow-on work.

1 Introduction

Spaceborne synthetic aperture radar (SAR) provides wide-area maritime observation independent of daylight and through most weather conditions [7]. Those properties make Sentinel-1 imagery useful for detecting vessels without a contemporaneous matched Automatic Identification System report—the so-called *dark vessels*. Such detections support maritime safety, fisheries monitoring, and investigation of activity potentially associated with sanctions evasion. The task is difficult: vessels occupy few pixels in a large scene, sea clutter and coastal structures produce strong responses, and human-verified labels are expensive. The xView3-SAR benchmark makes the problem concrete with ground-range-detected imagery, georeferenced point annotations, and an evaluation protocol based on geographic matching [3].

The scarcity of verified labels motivates transfer learning, but it is not clear which source helps most. Generic ImageNet pretraining supplies broad visual features; optical remote-sensing pretraining adds overhead-scene statistics; SAR pretraining additionally shares the sensing modality. These alternatives also interact with architecture: vision transformers and convolutional networks encode different inductive biases, and released remote-sensing checkpoints are not available under one method-matched recipe. A comparison that changes architecture, optimization, detector, and pretraining source simultaneously cannot isolate the source-domain question.

We study label efficiency through two architecture-matched tracks (Fig. 2, Appendix B). Within each track, four arms differ only in initialization: random, ImageNet, optical remote sensing, or SAR. Every arm fine-tunes end to end through the same point detector, input representation, optimizer, schedule, nested scene subsets, and fixed seed, producing 32 controlled cells. The study asks three questions:

1. How does random, ImageNet, optical, or SAR initialization change detection F1 as the number of labeled scenes grows?
2. Does SAR pretraining beat optical pretraining within each architecture, especially under label scarcity, and is the direction of that contrast consistent across tracks?
3. After model selection and thresholds freeze, how do the arms generalize to held-out scenes and human-verified dark vessels?

This report answers the first two questions with the completed 32-cell H100 campaign, every value re-derived at build time from hash-bound evidence. The 16-scene test matrix and the sealed 50-scene human-verified evaluation address the third and are described in Sect. 5. Our contribution is experimental control rather

than a new method: an eight-arm two-track design in which initialization is the only within-track variable, one corrected evaluation contract with checkpoint-bound operating thresholds, and a fail-closed evidence path from training bytes to every published number.

2 Dataset and Evaluation Protocol

Study cohort. We use xView3-SAR Sentinel-1 ground-range-detected scenes with VH and VV polarizations at 10 m pixel spacing [3]. Scene identifiers, not image chips, are the unit of partitioning. A frozen seed-0 cohort contains 150 study scenes—111 train, 23 development, and 16 test—stratified by coarse geographic region and shoreline presence; a separate set of 50 human-verified scenes is reserved for one sealed final evaluation. The four training budgets are exact nested sets of 12, 28, 56, and 111 scenes:

$$S_{10} \subset S_{25} \subset S_{50} \subset S_{100}. \quad (1)$$

Every arm sees the same scene set at a given budget, so a budget never changes which scenes one initialization sees relative to another. The preprocessing path converts each scene into a fixed three-channel decibel-domain tensor [VH, VV, VH − VV], computes normalization statistics from training scenes only, and cuts overlapping 800-pixel chips from which optimization samples shared 512-pixel crops.

Ground-truth contract. One target-semantics contract serves training and evaluation (Fig. 4, Appendix B): a row is a positive vessel point only when `is_vessel` is true at HIGH or MEDIUM confidence; HIGH/MEDIUM non-vessels are background; LOW-confidence rows define ignore regions. Under this contract the fixed eight-scene training-time development subset contains 517 positives, 107 background non-vessels, and 118 ignores, and the 16-scene test split contains 1,165 positives. These support counts are pre-registered integrity checks: evaluation stops if regenerated counts differ. Training never stages test or human-verified labels.

Selection and scoring. The frozen scorer greedily matches predictions to ground truth within 200 m and aggregates precision, recall, and F1 across scenes. Every five epochs, each run performs full-scene inference on the same eight development scenes and sweeps the score threshold that maximizes development F1; the best value is recorded with the retained checkpoint, and the protocol binds that complete operating point (epoch, threshold, and counts) to the SHA-256 of the exact checkpoint. Held-out stages apply each stored threshold verbatim—first to the 16 test scenes and then, once, to the human-verified set; held-out annotations never retune a threshold or select a checkpoint. The frozen split is leakage-free but not geographically independent: Sentinel-1 revisits place held-out scenes near training scenes, so we describe absolute performance as in-region generalization.

3 Related Work

xView3 frames dark-vessel detection as point localization in Sentinel-1 SAR rather than box classification [3]. A CenterNet-style heatmap fits the point annotations and distance metric without inventing box extents [9]; we use this established detector form to hold the task head fixed, and the detector is not presented as a contribution.

ImageNet pretraining is the common baseline for visual transfer learning [4]. The ViT track’s generic control is a supervised AugReg ViT-B/16 checkpoint; the CNN track’s is ConvNeXt-V2-Base initialized by fully convolutional masked autoencoding followed by supervised ImageNet-1K fine-tuning [8]. Remote-sensing pretraining may narrow the gap between source and target statistics: SatDINO applies DINO self-distillation to fMoW-RGB optical imagery [5], and SARMAE applies masked autoencoding to a million-scale SAR corpus [2]. Because these released ViT checkpoints differ in objective as well as sensing domain, their contrast is a best-available-checkpoint comparison rather than a pure modality intervention. The CNN track is tighter: its optical and SAR arms use BigEarthNet-v2 land-cover models trained with the same architecture, dataset family, and supervised task on Sentinel-2 and Sentinel-1 inputs, respectively [6, 1]—the study’s cleanest modality contrast. Matched roles across the two tracks then test whether a within-track pattern survives a change of architecture family.

Table 1: Controlled initialization roles. All non-random entries are released, revision-pinned checkpoints; task heads are discarded.

Role	ViT-B/16 track	ConvNeXt-V2-Base track
Random	fresh initialization	fresh initialization
Optical	SatDINO, DINO on fMoW-RGB	BigEarthNet-v2 Sentinel-2
SAR	SARMAE, MAE on SAR-1M	BigEarthNet-v2 Sentinel-1
ImageNet	supervised AugReg	FCMAE + supervised

4 Methods

Two matched tracks, eight arms. Table 1 lists the arms. Within each track the architecture and parameter count are invariant and only the initialization changes; pretrained checkpoints transfer the encoder only, with source heads discarded, and random floors use the identical target architecture. End to end, the implemented detectors contain 89,996,801 (ViT) and 93,988,865 (CNN) trainable parameters.

Shared point detector. Every arm predicts a stride-4 vessel-presence heatmap from the same three-channel input (Fig. 3, Appendix B). Gaussian targets with standard deviation two output pixels mark positives; LOW-confidence annotations define ignore neighborhoods. Training uses the penalty-reduced focal objective of CenterNet-style point detection [9]. The ViT adapter receives stride-16 features and the CNN adapter stride-32 features with one learned upsampling stage; both terminate in the same 256-channel head. At inference, local peaks above a permissive 0.05 candidate floor map to scene coordinates, 120 m distance non-maximum suppression removes duplicates, and whole scenes are tiled with 512-pixel windows at stride 384.

Uniform fine-tuning recipe. One PyTorch Lightning module, data module, and trainer configuration serve all cells. AdamW uses learning rate 10^{-4} , weight decay 0.05, layer-wise decay 0.65, five warm-up epochs, and cosine decay across at most 50 epochs; early stopping waits four development evaluations. Every cell uses micro-batch 16, accumulation 1, seed 0, and full-encoder fine-tuning. The campaign executed one process per NVIDIA H100 in Lightning 32-true with CUDA matmul and cuDNN TF32 disabled; the code itself is hardware-neutral, and the uniform hardware class is a property of the evidence. No hyperparameter is selected per arm or label fraction.

Registered contrasts. For track t , role r , and fraction f , let $F_{r,f}^{(t)}$ denote development F1 at that cell’s checkpoint-bound operating threshold. The analysis reports transfer over the track-matched random floor,

$$\Delta_{r,f}^{(t)} = F_{r,f}^{(t)} - F_{\text{random},f}^{(t)}, \quad (2)$$

and the within-track SAR-optical contrast,

$$\Delta_{\text{SAR-opt},f}^{(t)} = F_{\text{SAR},f}^{(t)} - F_{\text{optical},f}^{(t)}. \quad (3)$$

Agreement in the sign and pattern of Eq. (3) across tracks is the evidence used to assess architecture generality; it is never pooled into a single effect estimate. The evidence generator rejects a marker that does not hash to its cohort binding, a metric inconsistent with its own TP/FP/FN counts, a best-development value that disagrees with the training curve, a mixed hardware class, or a partial held-out cohort.

5 Results and Analysis

Execution summary and reporting contract. The core result is one complete cohort: eight arms at four nested budgets, seed 0, all 32 cells finished on one NVIDIA H100 80GB HBM3 class in strict FP32 (548.4 GPU-hours; code 1a82d508; cohort 9b1ba03e). Every published value is re-derived at build time from the committed evidence—the frozen cohort, hash-bound completion markers, and full training curves—and a single inconsistent byte stops the build. The reported metric is corrected development-selection F1:

Table 2: Development-selection F1 for all 32 cells at each cell’s checkpoint-bound operating threshold. Budgets are fractions of the 111 training scenes (scene counts in parentheses). Per-cell thresholds, best epochs, and held-out slots appear in Appendix A.

Track	Initialization	10% (12)	25% (28)	50% (56)	100% (111)
ViT-B/16	Random	0.8030	0.8563	0.8912	0.8958
	Optical	0.8902	0.8802	0.9136	0.9268
	SAR	0.8857	0.8624	0.9161	0.9289
	ImageNet	0.8761	0.8716	0.9284	0.9399
ConvNeXt-V2-B	Random	0.7992	0.8167	0.8664	0.8919
	Optical	0.7992	0.8063	0.8586	0.8636
	SAR	0.8762	0.8868	0.8998	0.9068
	ImageNet	0.8856	0.9005	0.9221	0.9387

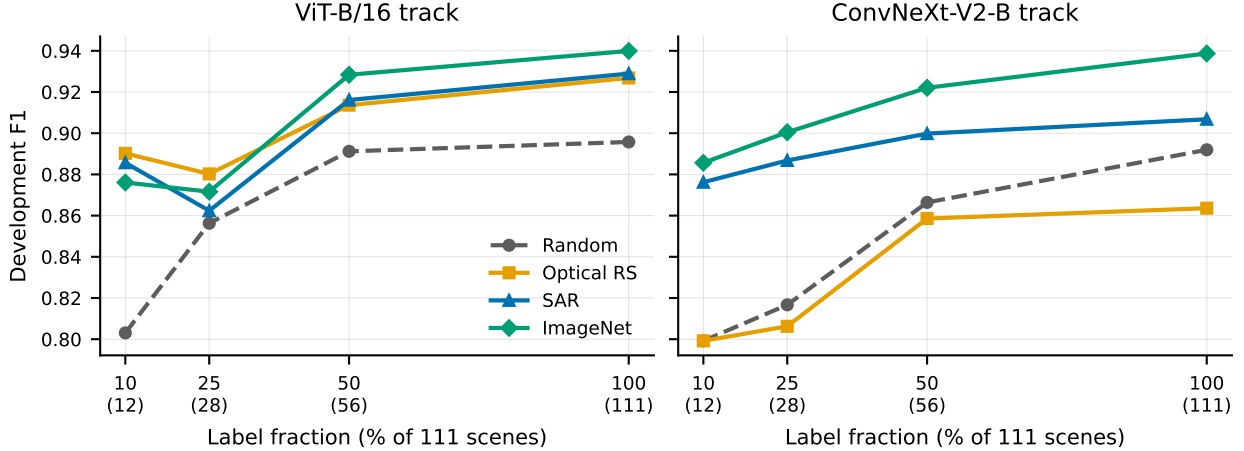


Figure 1: Development-F1 label-efficiency curves, one panel per track. Solid lines are pretrained arms; the dashed gray line is the track’s random floor. Values are seed-0 point estimates; no uncertainty ribbon is drawn.

each cell selects its checkpoint and operating threshold on the eight development scenes, so the values carry selection optimism and are not held-out estimates. Table 2 lists the matrix; Fig. 1 shows the four-point curves. Training dynamics and cost appear in Appendix B (Fig. 5).

Label-efficiency curves and transfer gains. The two tracks occupy different regimes. The ViT random floor is strong (0.8958 at full data), and every pretrained ViT arm clears it at 12 scenes: optical by +0.087, SAR by +0.083, and ImageNet by +0.073 (Eq. (2); gain panels in Fig. 6, Appendix B). The CNN track spreads wider: SAR gains +0.077 and ImageNet +0.086 over the 12-scene floor, while the optical Sentinel-2 arm only matches its floor at 12 scenes (both 0.7992) and falls below it at 25% and 50%—land-cover classification features that vessel detection must first unlearn.

Two curves violate monotonicity at the 10%–25% transition: ViT-optical drops from 0.8902 to 0.8802 and ViT-SAR from 0.8857 to 0.8624. We report these declines as observed rather than smoothing or rerunning them; the metric rests on a few hundred positives over eight fixed scenes, so a shift of a few hundredths corresponds to a handful of detections at the swept threshold. Held-out scoring on the sealed sixteen-scene split is the designed way to resolve such selection noise.

A discrete crossing summary asks for the smallest tested budget at which an arm reaches its track’s full-data random floor. CNN-ImageNet crosses at 28 scenes (0.9005 against 0.8919); every ViT transfer arm and CNN-SAR cross at 56 scenes; CNN-optical never crosses. No arm crosses at 12 scenes—the strongest 12-scene cell, ViT-optical at 0.8902, sits just below the ViT floor’s 0.8958—so under this development-selection metric, a well-chosen public checkpoint at 12 scenes nearly, but not fully, substitutes for 111 random-initialization

scenes.

Source-domain contrasts and architecture generality. The CNN track orders SAR above optical at every budget (Eq. (3); right panel of Fig. 6), with the largest contrast under scarcity (0.8762 versus 0.7992 at 12 scenes). Because the BigEarthNet Sentinel-1 and Sentinel-2 checkpoints share architecture, dataset family, and supervised task, this is the study’s cleanest modality effect. The ViT track disagrees below half data: optical leads SAR at 10% and 25% (0.8902 versus 0.8857; 0.8802 versus 0.8624), and SAR recovers only from 56 scenes upward. The sign of the SAR–optical contrast therefore does not replicate across architectures under label scarcity. A modality-matched encoder is not uniformly the best scarce-label choice; architecture and pretraining objective moderate the effect, and the released ViT checkpoints also differ in objective and source volume, so their contrast describes available checkpoints rather than a pure modality intervention.

At full data, generic ImageNet initialization leads both tracks (0.9399 and 0.9387): domain pretraining matters most where labels are scarce, and with enough target data generic visual structure plus full fine-tuning closes the domain gap.

Held-out results. The 16-scene test split had not been scored under the held-out contract when this report was built. The contract freezes the 32-cell cohort first, applies each cell’s development-bound threshold verbatim, and writes one immutable result per cell; the generator admits the test column only when all 32 results validate. Two cross-hardware control cells scored locally under this contract (RTX 5070 Ti, strict FP32) showed development-to-test changes of -0.080 and -0.069 , consistent with development-selection optimism rather than threshold-transfer failure. The 50 human-verified scenes remain sealed; dark-vessel recall and near-shore performance are defined only on that set and await its once-only evaluation.

6 Conclusion and Future Work

We built a controlled two-track experiment for measuring transfer under four nested label budgets: one detector, one corrected label contract, one strict-FP32 recipe, and a fail-closed path from training bytes to every published number. The completed 32-cell grid supports three conclusions. First, transfer concentrates its value under scarcity—at 12 scenes every pretrained arm except CNN-optical clears its random floor, and the best 12-scene cells approach what 111 scenes buy a randomly initialized detector. Second, matching the sensing modality is not uniformly the right scarce-label choice: SAR pretraining dominates optical in the CNN track at every budget, while the ViT track prefers the optical checkpoint below half data, so the contrast’s sign is architecture-dependent. Third, generic ImageNet initialization wins both tracks at full data. These remain corrected development-selection results from one seed and support no variance claim.

The next step scores the frozen 16-scene test split under the held-out contract, which applies each cell’s development-bound threshold verbatim. The 50 human-verified scenes then support one sealed evaluation of dark-vessel recall and near-shore performance for follow-on work.

Roles. John Roth and Kyle Wagner jointly designed the study, implemented its data, model, training, and evaluation code, ran the experiments, analyzed the evidence, and wrote this report.

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A Full Evidence Inventory

Table 3 lists every cell with its development-selection F1, checkpoint-bound operating threshold, epoch of the selected checkpoint, and held-out test F1 where admitted. All values are re-derived at build time from the committed cohort, markers, and training curves; dashes mean the fail-closed generator admitted no value, never zero performance. The recorded thresholds range widely (0.587–0.929 in the ViT track, 0.716–0.910 in the CNN track), confirming that a single shared threshold would misrepresent most arms and that the per-cell sweep does real calibration work.

Table 3: Per-cell evidence inventory (development-selection values; the test column is admitted only when all 32 immutable held-out results validate).

Cell	Budget	Dev F1	Threshold	Best epoch	Test F1
ViT random	10%	0.8030	0.822	19	—
ViT random	25%	0.8563	0.890	29	—
ViT random	50%	0.8912	0.853	44	—
ViT random	100%	0.8958	0.778	14	—
ViT optical	10%	0.8902	0.888	49	—
ViT optical	25%	0.8802	0.910	24	—
ViT optical	50%	0.9136	0.902	39	—
ViT optical	100%	0.9268	0.859	34	—
ViT SAR	10%	0.8857	0.587	4	—
ViT SAR	25%	0.8624	0.768	4	—
ViT SAR	50%	0.9161	0.855	39	—
ViT SAR	100%	0.9289	0.904	24	—
ViT ImageNet	10%	0.8761	0.906	39	—
ViT ImageNet	25%	0.8716	0.929	34	—
ViT ImageNet	50%	0.9284	0.844	49	—
ViT ImageNet	100%	0.9399	0.894	49	—
CNN random	10%	0.7992	0.716	29	—
CNN random	25%	0.8167	0.819	44	—
CNN random	50%	0.8664	0.857	44	—
CNN random	100%	0.8919	0.910	49	—
CNN optical	10%	0.7992	0.749	29	—
CNN optical	25%	0.8063	0.822	44	—
CNN optical	50%	0.8586	0.879	34	—
CNN optical	100%	0.8636	0.878	44	—
CNN SAR	10%	0.8762	0.776	34	—
CNN SAR	25%	0.8868	0.877	14	—
CNN SAR	50%	0.8998	0.893	24	—
CNN SAR	100%	0.9068	0.832	4	—
CNN ImageNet	10%	0.8856	0.869	4	—
CNN ImageNet	25%	0.9005	0.906	39	—
CNN ImageNet	50%	0.9221	0.899	19	—
CNN ImageNet	100%	0.9387	0.894	44	—

B Figures

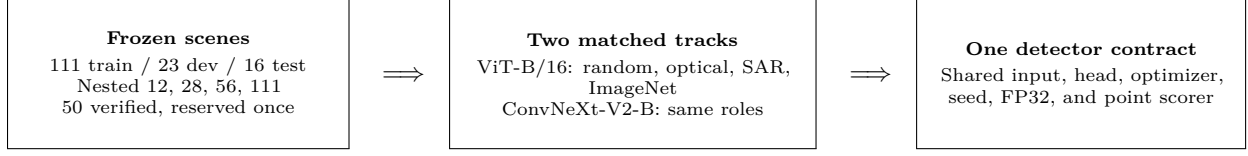


Figure 2: Controlled study design. Initialization is the only variable within each architecture track. Matched roles are compared descriptively across tracks; the final human-verified set remains sealed until the cohort and analysis code are frozen.

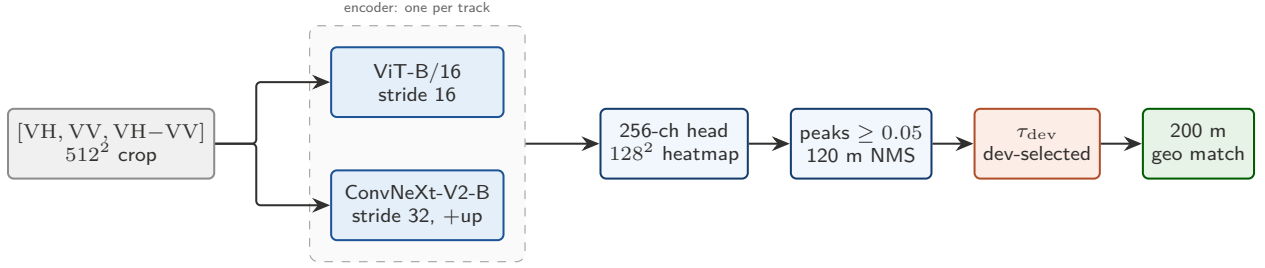


Figure 3: The shared detection path: only the encoder differs by track; one head, decode rule, dev-selected threshold, and frozen scorer serve every arm.

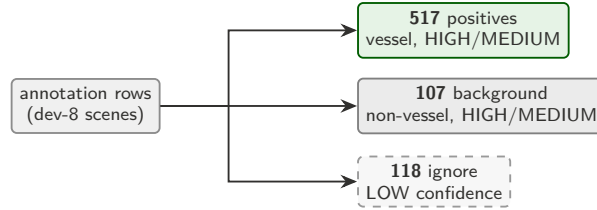


Figure 4: The corrected ground-truth contract on the training-time development subset. A row is a positive only when `is_vessel` is true at HIGH or MEDIUM confidence; HIGH/MEDIUM non-vessels are background, and LOW-confidence rows become ignore regions. Every H100 cell trained and evaluated under this contract.

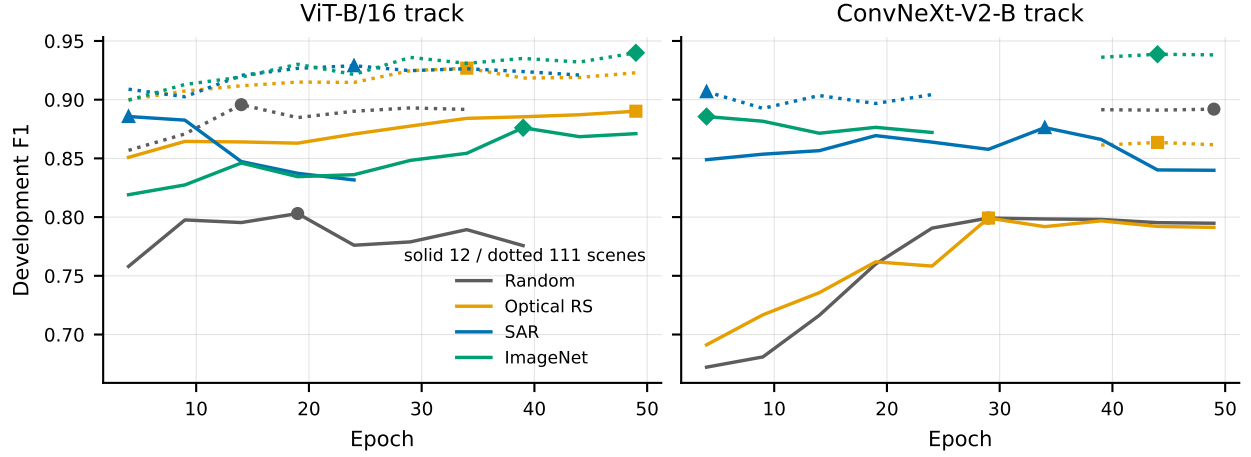


Figure 5: Development-F1 evaluation history for the smallest (12 scenes, solid) and largest (111 scenes, dotted) budgets, with each cell’s selected checkpoint marked. Pretrained CNN arms separate from the random floor within the first evaluations; several strong arms peak early and are retained by the checkpoint binding.

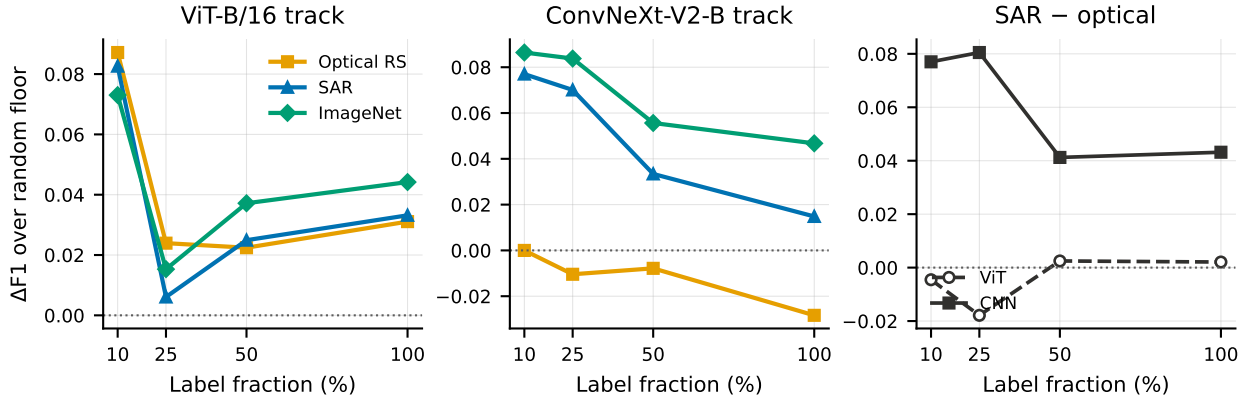


Figure 6: Left, center: transfer gain over the track-matched random floor (Eq. (2)) at each budget; the optical CNN arm is at or below zero everywhere. Right: the SAR-optical contrast (Eq. (3)); its sign flips between tracks below half data, so the modality effect is architecture-dependent.

C Reproducibility and Audit Record

The evidence tree records the frozen training cohort, each cell’s hash-bound completion marker, its full training curve, and a sanitized runtime receipt (H100 hardware class, source SHA, strict-FP32 backend state). The generator validates marker-to-cohort hashes, metric consistency, and curve agreement before writing a manuscript artifact. Checkpoint bytes stay outside the repository; their SHA-256 bindings are published for archive verification. The submission excludes imagery, labels, weights, checkpoints, run trees, credentials, and private paths.

The detector uses 512-pixel crops, a stride-4 heatmap, Gaussian standard deviation two output pixels, a candidate floor of 0.05, and 120m distance NMS. Training uses at most 50 epochs with development evaluation every five epochs. The project fixes seed 0 and reports point estimates without uncertainty bands.

D Qualitative-Evidence Boundary

The committed evidence contains metrics, curves, and receipts rather than exported scene overlays. We omit qualitative examples instead of substituting engineering probes or opening sealed imagery. After the held-out results land through the fail-closed path, we will sample successes and errors from the pre-registered held-out outputs.